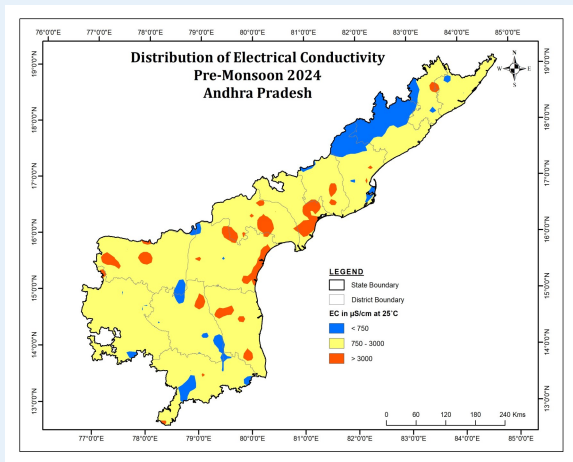
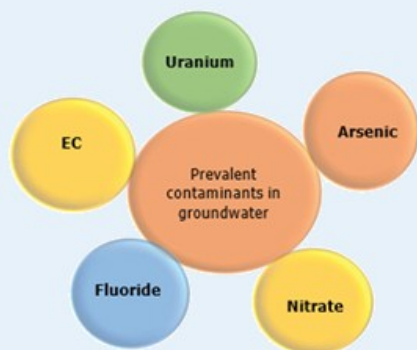


Groundwater Quality Scenario in Andhra Pradesh

Parameters	No of samples	Permissible limit	No of Samples above permissible limit	% Samples above permissible Limit
EC	1135	3000 $\mu\text{S}/\text{cm}$	111	9.78
Fluoride	1135	1.5 mg/L	102	8.99
Nitrate	1135	45 mg/L	275	24.23



Districts with anomalous values at sporadic locations

EC (3000 $\mu\text{S}/\text{cm}$)

Anakapalli, Anantapuram, Annamayya, Bapatla, Chittoor, East Godavari, Eluru, Guntur, Kakinada, Konaseema, Krishna, Kurnool, Nellore, NTR, Palnadu, Parvathipuram Manyam, Prakasham, Sri Satya Sai, Srikakulam, Tirupati, Vizianagaram, West Godavari, YSR Kadapa.

Fluoride (F > 1.5 mg/L)

Alluri Sita Rama Raju, Anakapalli, Anantapuram, Annamayya, Bapatla, Chittoor, Krishna, Kurnool, Nandyal, Nellore, NTR, Palnadu, Prakasham, Sri Satya Sai, Tirupati, YSR Kadapa.

Nitrate (Nitrate > 45 mg/L)

Alluri Sita Rama Raju, Anakapalli, Anantapuram, Annamayya, Bapatla, Chittoor, East Godavari, Eluru, Guntur, Kakinada, Konaseema, Krishna, Kurnool, Manyam, Nandyal, Nellore, Ntr, Palnadu, Parvathipuram Manyam, Prakasham, Sri Satya Sai, Srikakulam, Tirupati, Visakhapatnam, Vizianagaram, West Godavari, YSR Kadapa.

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Email: chmn-cgwb@nic.in



<https://ingres.iitb.ac.in>

<https://cgwb.gov.in>

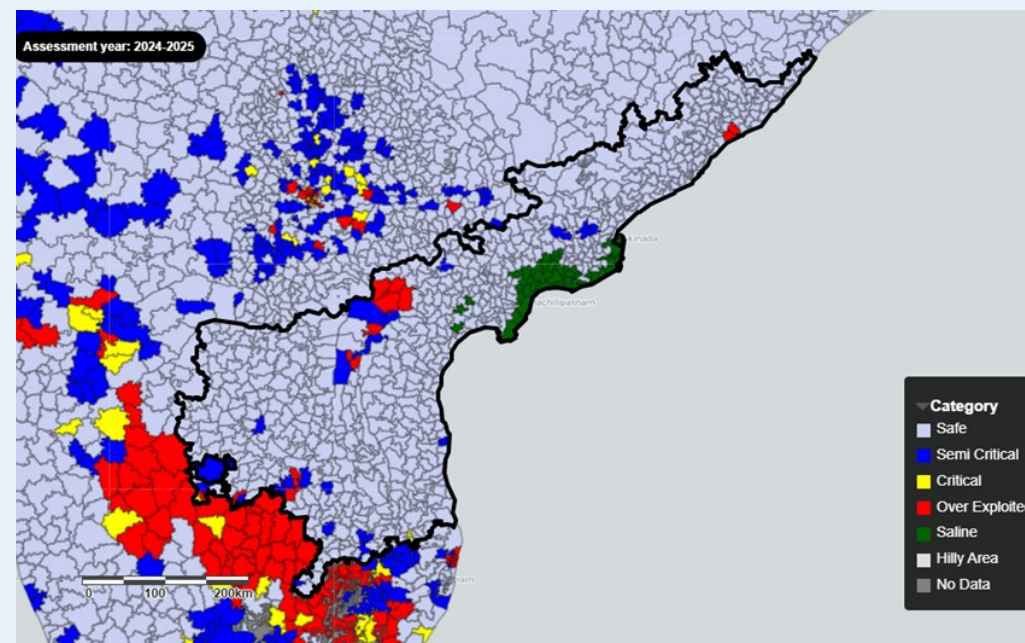
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Central Ground Water Board Department of Water Resources, RD & GR Ministry of Jal Shakti, Government of India



Dynamic Ground Water Resources, Ground Water Level & Ground Water Quality of Andhra Pradesh, 2025

November, 2025

Groundwater Resource Scenario in Andhra Pradesh

- Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Mandal).
- The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- GWRA - 2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad.

Salient Features

1	Rainfall	891.99 mm
2	Hydrogeology	Nearly 80 % of the State is underlain by Hardrock formations (Granites, gneisses, metamorphic and intrusive (Archaeans), Pre-cambrian quartzites, shales and limestones (Cuddapahs & Kurnoos), Mesozoic Deccan Trap basalts etc.). Rest of the State is underlain by Gondwana, Tertiary sedimentaries and Sub Recent-Recent alluvium.
3	Recharge Worthy Area of the State	1.40 Lakh Sq. Km
4	Assessment Unit (AU) Type / Number	Mandal / 679 Numbers
5	Average area of Assessment Unit	206. 19 Sq. Km

Findings

	Attribute	GWRA-2017	GWRA-2020	GWRA-2022	GWRA-2023	GWRA-2024	GWRA-2025
1	Total Annual Ground Water Recharge (in bcm)	21.22	24.15	27.23	27.83	27.80	26.34
2	Annual Extractable Ground Water Resources (in bcm)	20.15	22.94	25.86	26.45	26.41	25.02
3	Annual Ground Water Extraction (in bcm)	8.90	7.63	7.45	7.48	7.88	7.88
4	Stage of Ground Water Extraction (in %)	44.15	33.26	28.81	28.30	29.83	31.51

bcm: Billion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

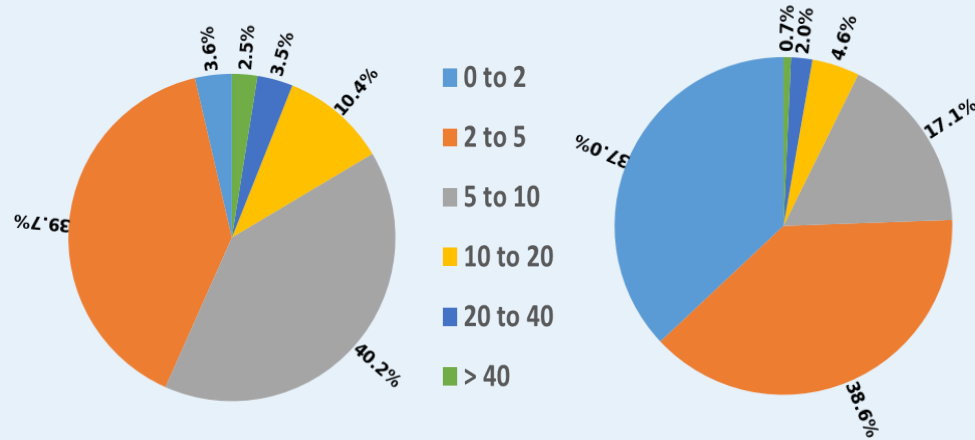
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs
1	Safe	501	75	551	83	598	90	597	90	591	87.04	601	88.50
2	Semi-critical	60	9	40	6	19	2.69	18	3	38	5.60	24	3.50
3	Critical	24	4	15	2	5	0.7	3	0.4	2	0.29	3	0.4
4	Over-exploited	45	7	23	3	6	0.9	10	1.5	9	1.33	12	1.8
5	Saline	40	6	38	6	39	6	39	6	39	5.74	39	5.74
Total number of AUs		670		667		667		667		679		679	

Recommendations: -

- Major parts (about 80%) of the state Andhra Pradesh is occupied by the hard rock formation. Ground water conditions in hard rocks are generally considering to be poor. The total annual groundwater recharge for the state is estimated at 26.34 bcm, with an annual extractable resource of 25.02 bcm. The current annual groundwater extraction for all uses is 7.88 bcm, and the stage of groundwater extraction is 31.51 %. Out of 679 assessment units (mandals), 12 (1.80%) units have been categorized, as 'Over-exploited', 03 units (0.4 %) as 'Critical', 24 units (3.50 %) as 'Semi-Critical', 601 units (88.50 %) as 'Safe' and 39 units categorized as 'Saline' (5.74%)
- More numbers of Water Harvesting and Conservation Structures may be constructed to catch the rain as the State is blessed with more than 800 mm annual rainfall particularly in the hard rock terrain. State may also effectively use "Master plan for Artificial Recharge" prepared by CGWB in consultation with State Government (<https://cgwb.gov.in/cgwbpnml/publication-detail/324>).
- Restoration/rejuvenation of all the existing tanks should be taken up with the view of accommodating the available surface run off and thus augmentation of the ground water resources by artificial recharge. Periodical maintenance of these tanks is to be ensured. The "Manual on Artificial Recharge Techniques for augmentation of ground water" prepared by CGWB may be used for planning (<https://cgwb.gov.in/sites/default/files/MainLinks/Manual-Artificial-Recharge.pdf>).
- National Aquifer Mapping & Management Programme (NAQIM) Reports prepared by CGWB (<https://cgwb.gov.in/cgwbpnml>) which are also being shared with State/District Authorities and Ground Water Year Book published by CGWB having water level & water quality data may be used in Ground water management (<https://cgwb.gov.in/cgwbpnml>).
- Increase in irrigation efficiency through adopting of micro-irrigation techniques in more areas.
- State may review their free/subsidized electricity policy to farmers (if applicable), bring suitable water pricing policy and may work further towards crop rotation/diversification/other initiatives to reduce overdependence on groundwater. In coastal and "Safe" Category of assessment areas of Andhra Pradesh, State Government can judiciously develop the ground water resource mainly for agricultural use, however, at no point of time the extraction level should exceed 70% .
- Creating awareness (Mass Awareness Campaign for public and farmers, Slideshows, Display boards on water conservation, Water Management Training Programme for personnel related with water sector, Painting/Essay competition for school students etc.) regarding water conservation etc may be organized at appropriate level.
- Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in Andhra Pradesh

1. Depth to Groundwater Level

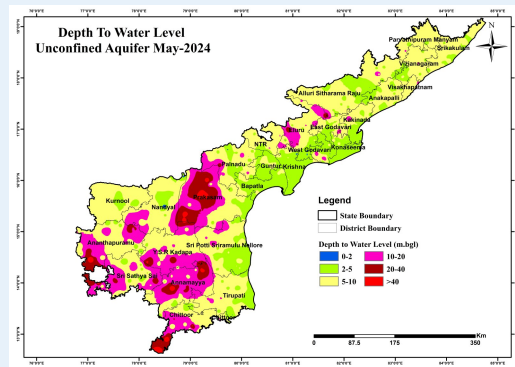
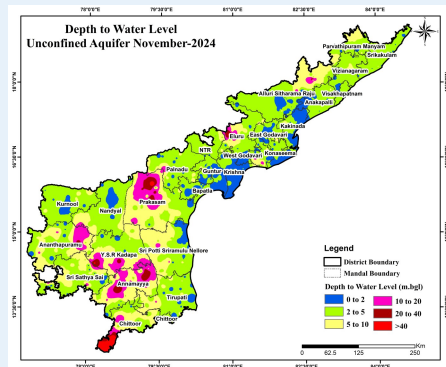


Pre-Monsoon (May 2024)

Analysis of 775 wells shows that groundwater depths mostly range from 2–10 m bgl (over 80% of wells and ~74% of the state), with shallow levels confined to small coastal and isolated patches and deeper levels (>20 m) limited to about 7% of the state in select inland districts.

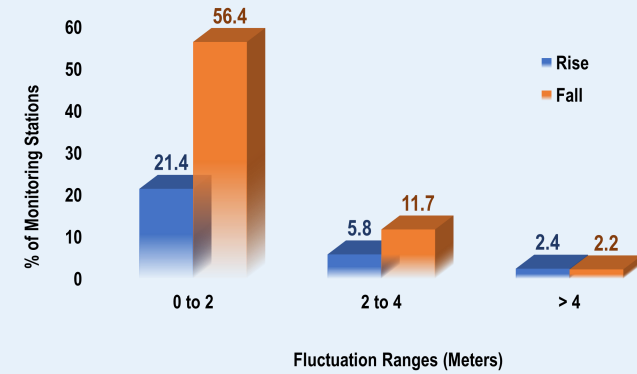
Post-Monsoon (November 2024)

About 75% of wells recorded groundwater levels within 5 m bgl, while around 25% showed depths greater than 5 m bgl, indicating generally favorable post-monsoon groundwater conditions with localized stress areas in the state.



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

